

# SIMATS ENGINEERING

## DSA0603 — DATA HANDLING AND VISUALIZATION

### CO3\_AT2\_ASSESSMENT TOOL II : DATA ANALYSIS PROJECT

|                                                |                         |                                                         |
|------------------------------------------------|-------------------------|---------------------------------------------------------|
| <b>Course: Data Handling and Visualization</b> | <b>Code: DSA0603</b>    | <b>CO: CO3</b>                                          |
| <b>Duration: 180 min</b>                       | <b>Total Marks: 100</b> | <b>Reg No: 192424281</b><br><b>Name: DILEEP KUMAR.D</b> |

#### COURSE OUTCOME COVERED

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods. (BL4)

Activity Tool : Data Analysis Project

Weightage: 20%

#### Code of Conduct:

I (DILEEP KUMAR DESHABOINA/192424281)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *DILEEP KUMAR.D*

**Topics Covered:** ECDF and Q-Q Plots; Multiple Distributions; Vertical-Axis Distributions; Bean Plots; Proportions; Pie/Side-by-Side/Stacked Bars; Stacked Densities; Treemaps; Sunburst Charts; Parallel Sets; Sankey Diagrams; Nested Proportions; Avoiding Misleading Proportions.

**General Instruction:** For each project task, use an appropriate visualization in R, justify the design choice, interpret the result in context, and identify potential visualization bias. R/ggplot2 (or an equivalent tool such as treemapify, ggalluvial, or networkD3) may be used. Submit R code, output plots, and a short written interpretation for each question.

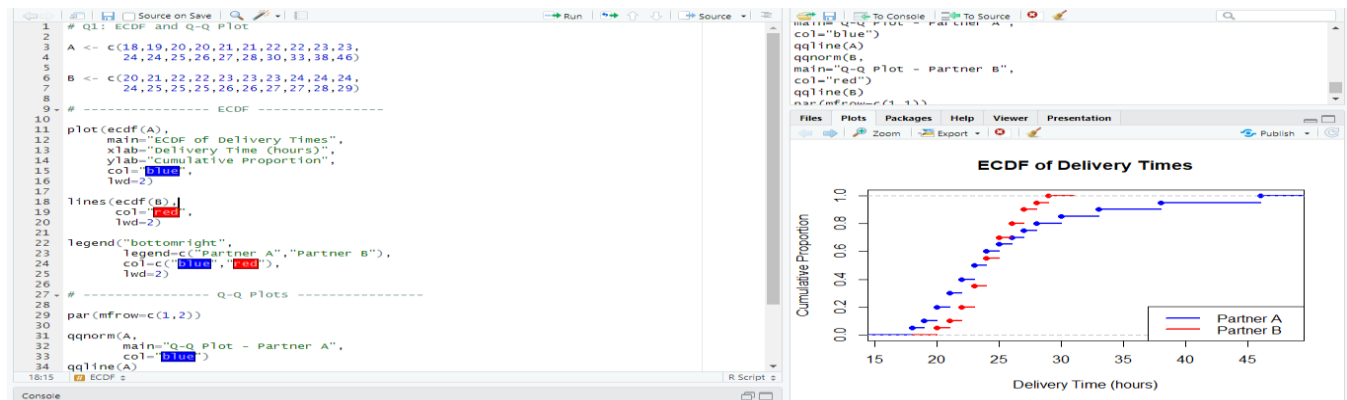
| Criteria | Understanding of Concepts 25% | Analytical Design 25% | Use of Tools 20% | Analysis of Results 20% | Documentation 10% | Total |
|----------|-------------------------------|-----------------------|------------------|-------------------------|-------------------|-------|
| Q1       |                               |                       |                  |                         |                   |       |
| Q2       |                               |                       |                  |                         |                   |       |
| Q3       |                               |                       |                  |                         |                   |       |
| Q4       |                               |                       |                  |                         |                   |       |
| Q5       |                               |                       |                  |                         |                   |       |

# Question 1 — Comparing Distributions with ECDF and Q-Q Plots

**Scenario:** A logistics company records delivery time (hours) for two courier partners over 20 shipments each. Operations wants to know whether one partner is consistently faster and whether either partner's delivery times follow a normal pattern that would justify simple mean-based SLAs.

**Dataset:**

| Partner   | Delivery Times (hours), n = 20                                                 |
|-----------|--------------------------------------------------------------------------------|
| Partner A | 18, 19, 20, 20, 21, 21, 22, 22, 23, 23, 24, 24, 25, 26, 27, 28, 30, 33, 38, 46 |
| Partner B | 20, 21, 22, 22, 23, 23, 23, 24, 24, 24, 24, 25, 25, 25, 26, 26, 27, 27, 28, 29 |



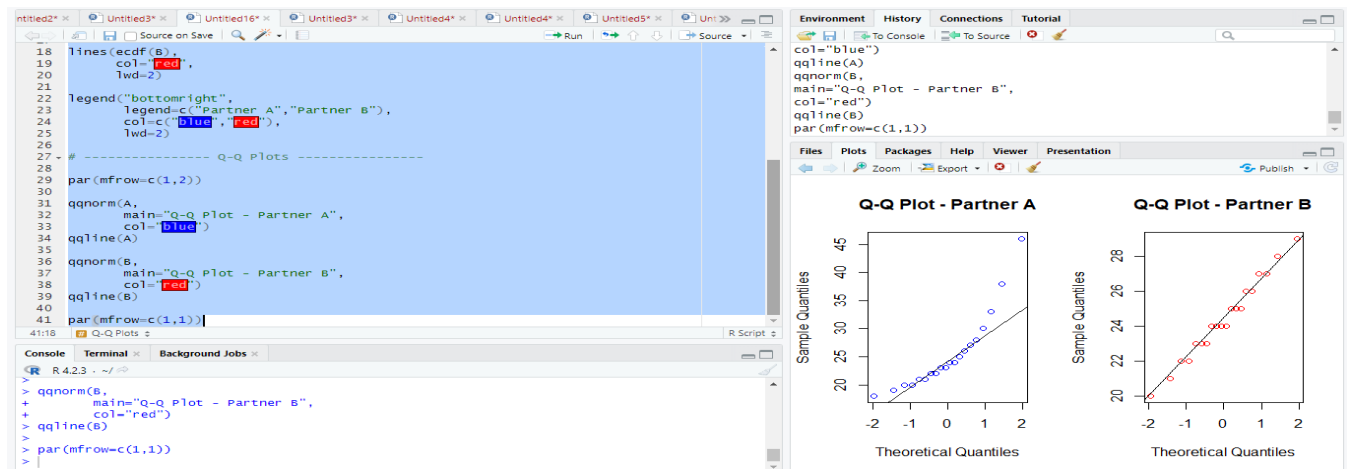
## 1. ECDF interpretation

The ECDF curves show that **Partner A is generally faster at lower delivery-time thresholds.**

The maximum difference occurs around **21–22 hours**, where the ECDFs differ by about **0.20 (20 percentage points)**.

At 21 hours:

- Partner A: 30% completed
- Partner B: 10% completed



## 2. Q-Q plot interpretation

**Partner A** has strong deviations from the straight line at the upper end because of delivery times such as **38 and 46 hours**. This indicates **right skewness and possible high-end outliers**.

**Partner B** follows the normal line more closely, although some deviations remain.

### 3. SLA recommendation

**Empirical percentiles are preferable to mean  $\pm$  standard deviation.**

Reason:

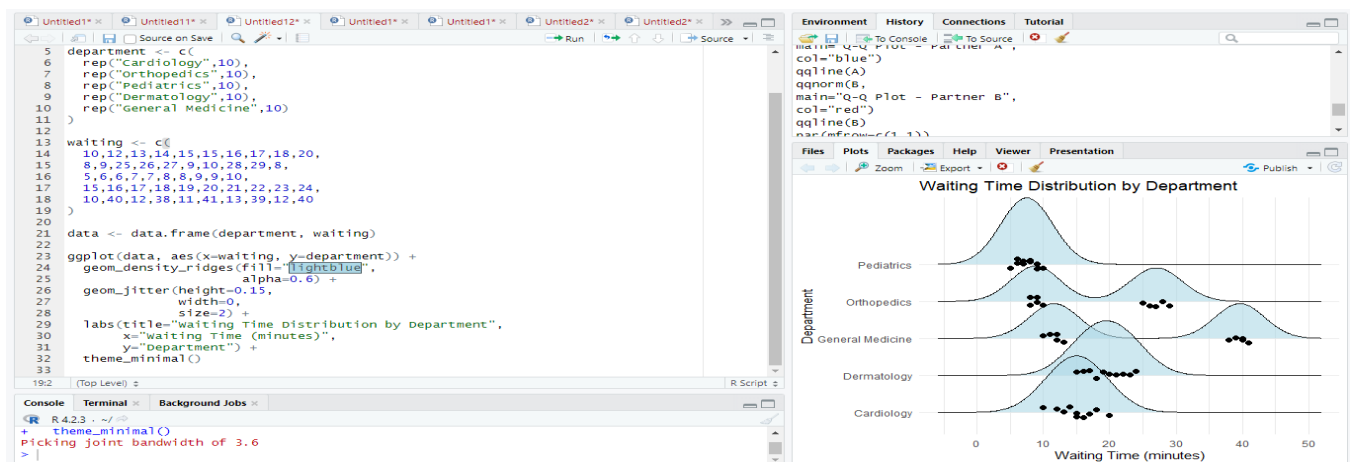
- Partner A has extreme high delivery times.
- The distribution is not perfectly normal.
- Mean and standard deviation can be strongly affected by outliers.
- Percentiles such as P90 or P95 directly represent actual delivery performance.

**Recommended SLA:** Use an empirical percentile such as the **90th or 95th percentile**.

## Question 2 — Distribution Shape Across Many Groups using Bean Plots

departments over a week. Administration wants to identify which departments have consistent waiting times and which have unpredictable, multi-modal patterns before reallocating staff.**Scenario:** A hospital network measures patient waiting time (minutes) in five outpatient

| Department       | Waiting Times (minutes)                |
|------------------|----------------------------------------|
| Cardiology       | 10, 12, 13, 14, 15, 15, 16, 17, 18, 20 |
| Orthopedics      | 8, 9, 25, 26, 27, 9, 10, 28, 29, 8     |
| Pediatrics       | 5, 6, 6, 7, 7, 8, 8, 9, 9, 10          |
| Dermatology      | 15, 16, 17, 18, 19, 20, 21, 22, 23, 24 |
| General Medicine | 10, 40, 12, 38, 11, 41, 13, 39, 12, 40 |



**Interpretation**

**General Medicine** clearly shows a **bimodal distribution**.

- routine patients
- complicated/emergency patients
- **Orthopedics** also shows two groups:

### Variability ranking

From highest to lowest variability:

1. **General Medicine**
2. **Orthopedics**
3. **Dermatology**
4. **Cardiology**
5. **Pediatrics**

### Median ranking

From highest to lowest median:

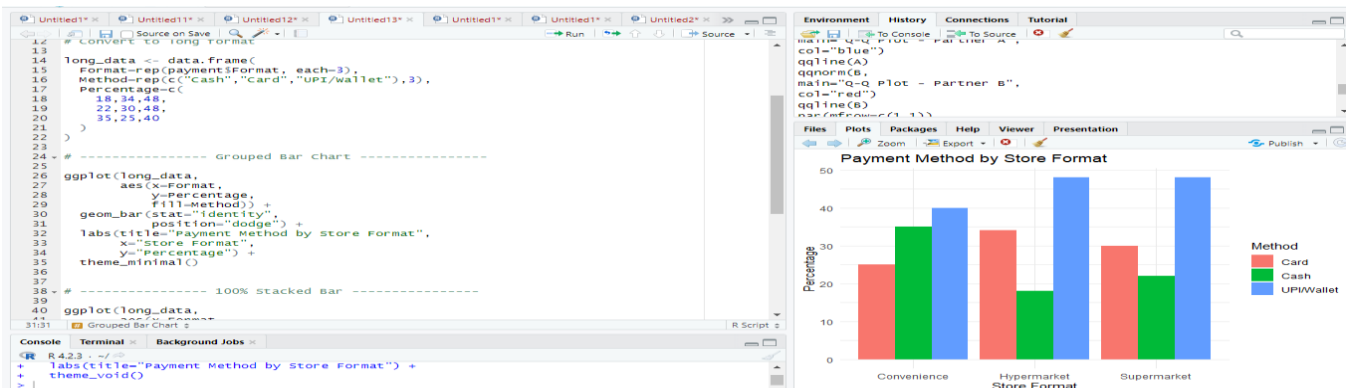
1. **General Medicine – 25.5**
2. **Dermatology – 19.5**
3. **Orthopedics – 17.5**
4. **Cardiology – 15**
5. **Pediatrics – 7.5**

### Question 3 — Choosing an Honest Proportion Chart

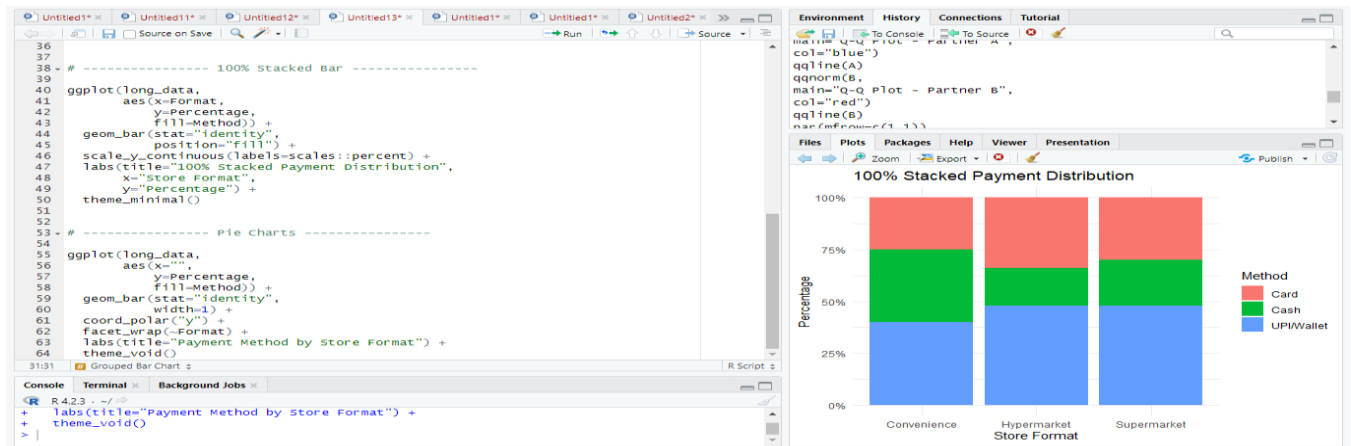
**Scenario:** A retail chain surveys 300 customers across three store formats about preferred payment method. Senior management wants a dashboard chart that allows accurate comparison of payment-method share across formats without visually exaggerating small gaps.

| Store Format | Cash (%) | Card (%) | UPI/Wallet (%) |
|--------------|----------|----------|----------------|
| Hypermarket  | 18       | 34       | 48             |
| Supermarket  | 22       | 30       | 48             |
| Convenience  | 35       | 25       | 40             |

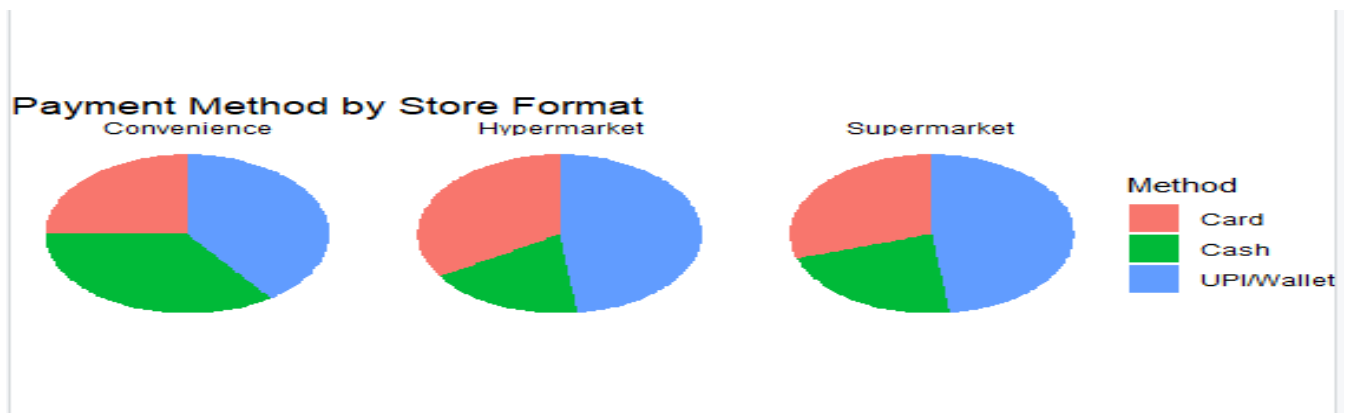
### Grouped Bar Chart



## Stacked Bar



## Pie Charts



## Comparison

The **grouped bar chart** is the **easiest** for comparing UPI/Wallet percentages.

UPI/Wallet:

**Format**    **UPI/Wallet**

Hypermarket 48%

Supermarket 48%

Convenience 40%

**Why pie charts are harder**

Bar charts use **length**, which humans compare accurately.

Pie charts use **angles and arc lengths**, which are harder to compare precisely.

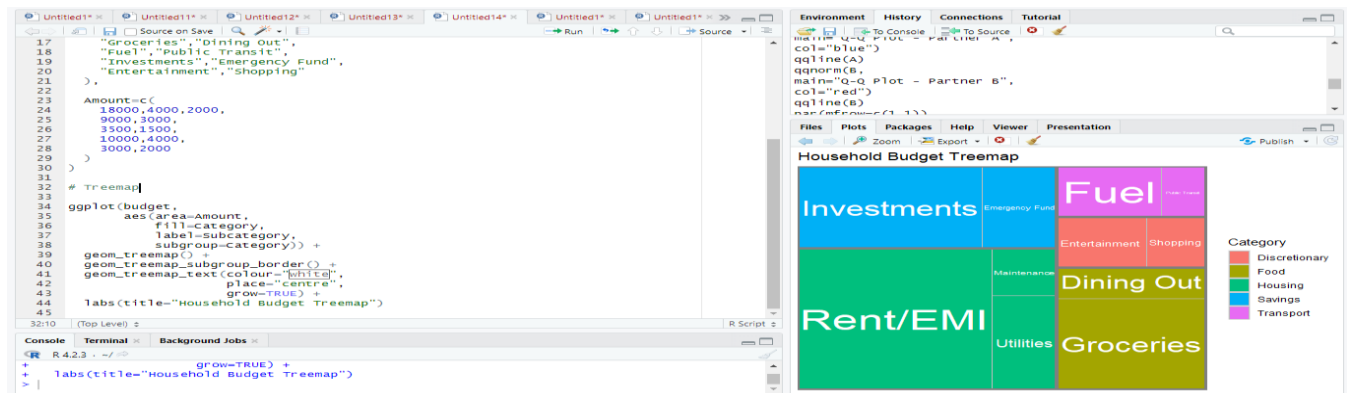
- 48%
- 40%

## Question 4 — Nested Proportions in a Household Budget

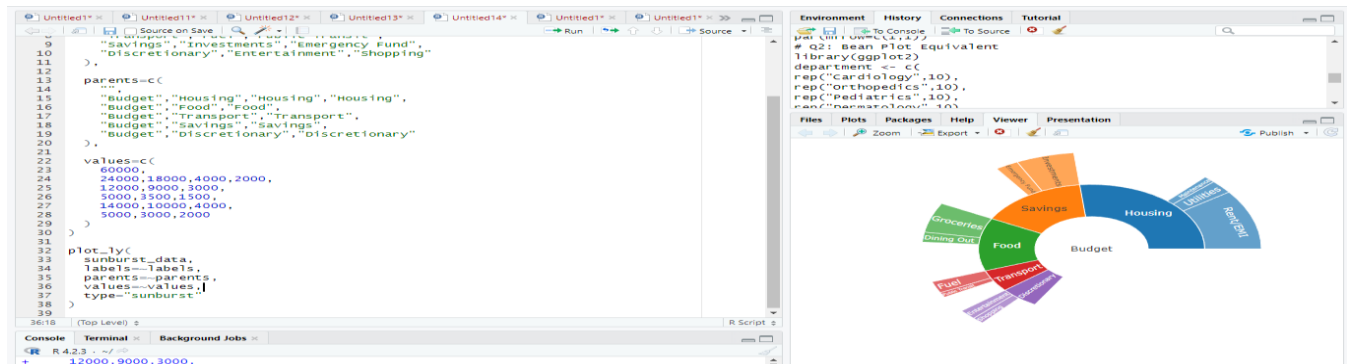
**Scenario:** A financial-planning app aggregates the average monthly household budget (₹60,000) across spending categories and subcategories for a client segment. The product team wants a dashboard visual that shows both overall category share and subcategory composition within each category.

| Category      | Subcategory    | Amount (₹) |
|---------------|----------------|------------|
| Housing       | Rent/EMI       | 18000      |
| Housing       | Utilities      | 4000       |
| Housing       | Maintenance    | 2000       |
| Food          | Groceries      | 9000       |
| Food          | Dining Out     | 3000       |
| Transport     | Fuel           | 3500       |
| Transport     | Public Transit | 1500       |
| Savings       | Investments    | 10000      |
| Savings       | Emergency Fund | 4000       |
| Discretionary | Entertainment  | 3000       |
| Discretionary | Shopping       | 2000       |

### Treemap



### Sunburst Chart



### Housing percentage

Housing:

₹18,000 + ₹4,000 + ₹2,000

= ₹24,000

### Treemap vs Sunburst

#### Treemap:

- Uses rectangular area.
- Good for seeing relative size.
- Efficient use of screen space.

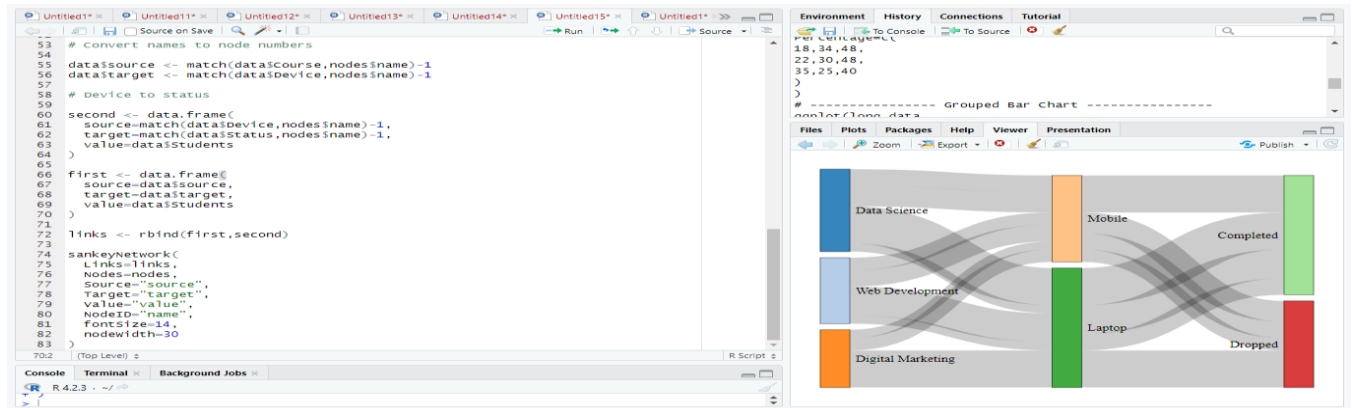
#### Sunburst:

- Clearly shows hierarchy.
- Category → subcategory relationship is intuitive.
- Comparing subcategories from different categories can be difficult because they appear at different angular positions/rings.

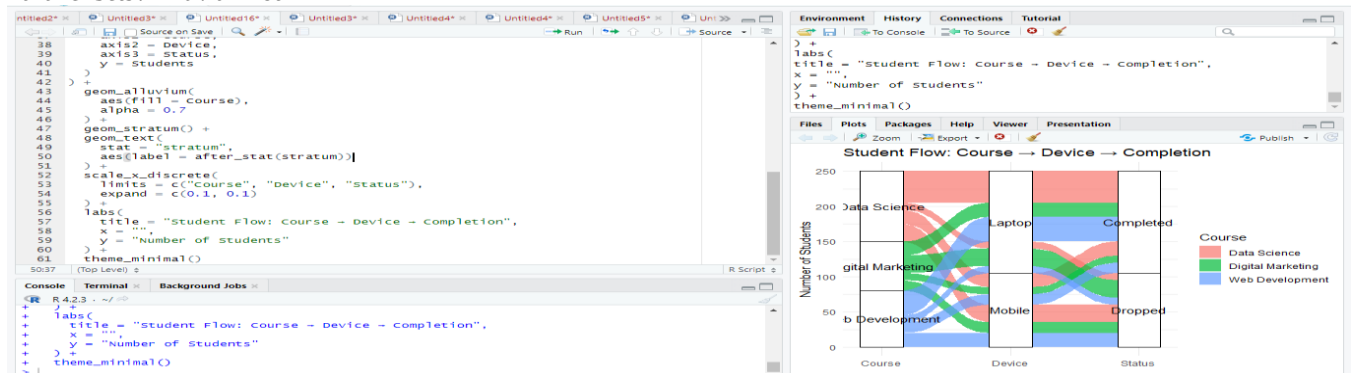
### Question 5 — Flow Analysis using Sankey Diagram and Parallel Sets

**Scenario:** An online learning platform tracks 250 students from the course they enrolled in, through the device used to study, to their final completion status. The academic team wants to understand dominant enrollment-to-outcome pathways and spot problem areas.

| Course            | Device | Completion Status | Students |
|-------------------|--------|-------------------|----------|
| Data Science      | Mobile | Completed         | 20       |
| Data Science      | Mobile | Dropped           | 25       |
| Data Science      | Laptop | Completed         | 45       |
| Data Science      | Laptop | Dropped           | 10       |
| Web Development   | Mobile | Completed         | 15       |
| Web Development   | Mobile | Dropped           | 20       |
| Web Development   | Laptop | Completed         | 35       |
| Web Development   | Laptop | Dropped           | 10       |
| Digital Marketing | Mobile | Completed         | 10       |
| Digital Marketing | Mobile | Dropped           | 15       |
| Digital Marketing | Laptop | Completed         | 20       |
| Digital Marketing | Laptop | Dropped           | 25       |



## Parallel Sets / Alluvial Plot



## Three largest flows

From the dataset:

| Flow                                 | Students | Percentage |
|--------------------------------------|----------|------------|
| Data Science → Laptop → Completed    | 45       | 18%        |
| Web Development → Laptop → Completed | 35       | 14%        |
| Data Science → Mobile → Dropped      | 25       | 10%        |

There is a **tie at 25 students** for some other flows, including Digital Marketing → Laptop → Dropped.

For example:

$$25045 \times 100 = 18\% \quad 25035 \times 100 = 14\% \quad 25025 \times 100 = 10\%$$

**Course with weakest completion**

**Data Science**